Independent validation of Softmax numerical repairs

Research validation report

2026-10-31

Summary

Softmax arithmetic can lose scientifically relevant derivatives while conventional stability checks still pass. We evaluate the published repairs [1] using independently defined high-precision references, deliberately broken controls and saved teacher–student models. The tested kernels preserve small target gradients and exact stationarity within the qualified input range. The training experiment separates numerical correctness from generalisation. This report records the repair, unfavorable outcomes and reproducible artifacts, with explicit limits on backend qualification and downstream claims.

Scope and research gap

Source: A Fix for Softmax and Cross-Entropy, v3. This reproduces published numerical mechanisms and repairs the existing Paper2Agent integration. It is not a novel scientific discovery or a reproduction of the paper's training outcomes.

The scientific gap in the source paper is a numerical failure hidden by conventional stability checks: a finite, well-scaled loss can still have an incorrect gradient. Eidos uses this as an adversarial acceptance case: reproducing a displayed number alone is insufficient.

Mechanisms

For hard target 0 and $z = (m, 0, 0)$, the exact target derivative is $-2e^{-m}/(1+2e^{-m})$. The standard subtraction `q[0]-1` becomes zero after the precision cliff while the other derivatives remain positive. A max shift prevents overflow but does not repair this subtraction. The fixed forward excludes exactly one argmax before `log1p`; the hard-target backward computes the target derivative as the negative sum of the other derivatives (source Eqs. 5-7).

For distillation, the true gradient is student probability minus teacher probability. Near equal logits this

subtraction loses relative accuracy; exact equality should yield zero. The repair uses centered logit differences, `expm1`/`log1p`, and a zero-sum correction, with the ordinary difference retained for the far branch (source Eqs. 8-9). Teacher logits are required for this contract; already-rounded probabilities cannot recover lost information.

Methods

Numerical execution and independent references

Backend: numpy 2.5.4. Decimal references use represented inputs and adaptive precision. Eidos 0.4.4; Python 3.12.4. No random seed applies to this deterministic grid. Source and lock SHA-256 hashes are recorded in evidence.json; bundle checksums cover the shipped source. The browser selects recorded cases; it does not simulate float32 training.

Statistics and experimental units

The 96 numerical cases are deterministic input settings, not 96 independent random replicates. No error bars, confidence intervals or significance tests are used for this grid. The teacher–student experiment reports every one of three declared seed pairs; an experimental unit is one trained model, and the paired arms share their initial teacher and split. Individual trajectories are shown.

Numerical results

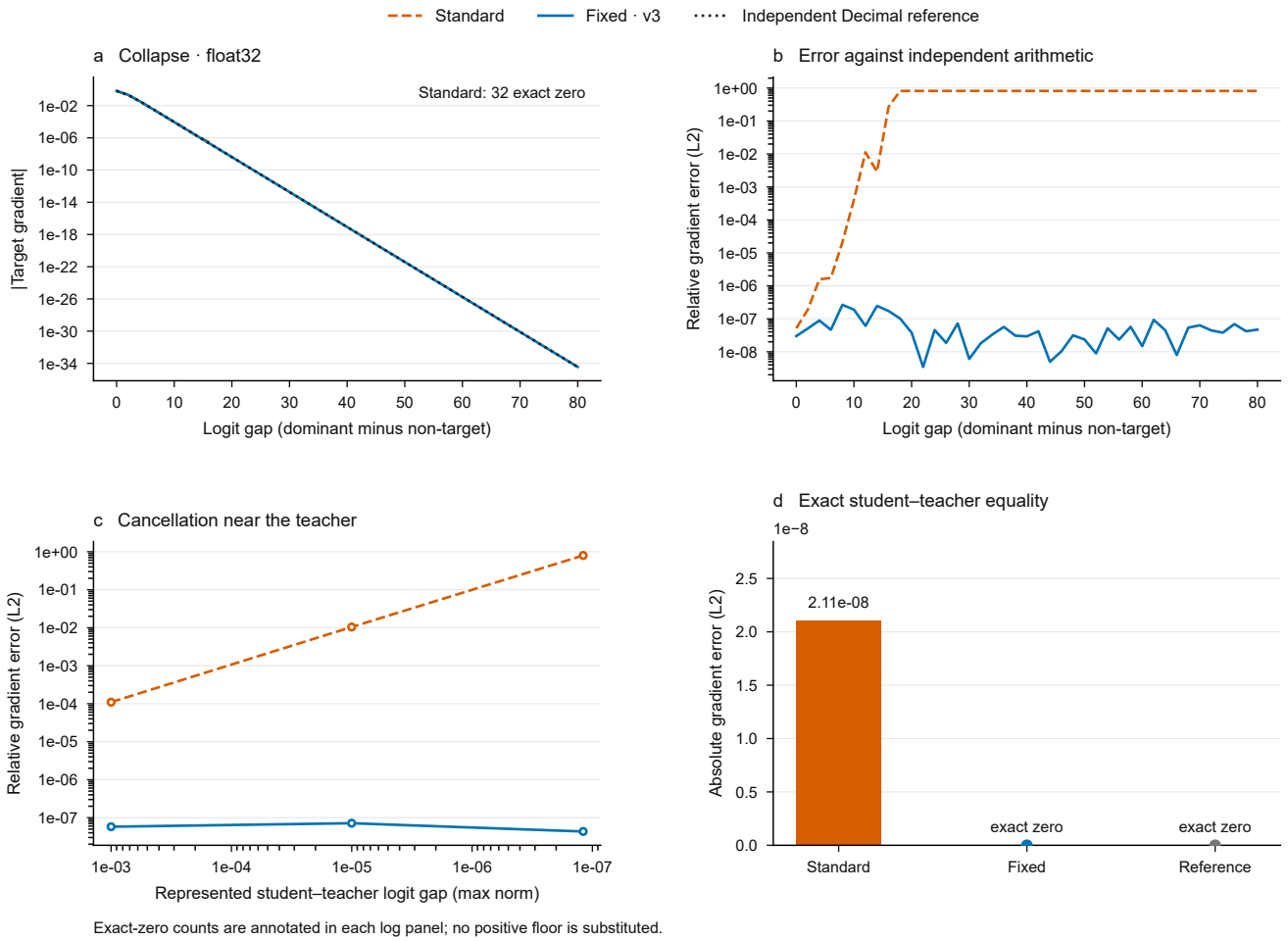


Figure 1 | Numerical repair in float32. a, Target-gradient magnitude across the logit gap. b, Relative L2 gradient error against independent Decimal arithmetic. c, Relative error near the teacher, using represented logit differences. d, Absolute L2 error at exact teacher equality. Lines show deterministic cases; no error bars or random sampling apply. Exact zeros are labelled and excluded from logarithmic traces. All quantities are dimensionless.

Table 1 | Selected collapse cases and dimensionless gradient errors.

Input dtype	Dominant logit	Standard target gradient	Fixed target gradient	Fixed relative L2 error
float32	18	0.000000e+00	-3.045996e-08	1.019e-07
float64	40	0.000000e+00	-8.496709e-18	2.078e-17
float64	80	0.000000e+00	-3.609703e-35	7.001e-17

Cancellation and stationarity

Table 2 | Requested and represented perturbations and absolute gradient errors.

Input dtype	Requested perturbation	Represented perturbation	Standard absolute error	Fixed absolute error
float32	0.0e+00	0.000000e+00	2.107342e-08	0.000000e+00
float32	1.0e-07	-1.192093e-07	4.950131e-08	2.682887e-15
float64	0.0e+00	0.000000e+00	2.775558e-17	0.000000e+00
float64	1.0e-07	-1.000000e-07	3.861881e-17	7.080722e-25

Controls and acceptance

Qualification: **PASS** on 12 cases. Both deliberately broken controls must be rejected: a lost target gradient and nonzero stationary gradient. Validation uses relative gradient errors, not an absolute tolerance floor.

What the Eidos pipeline demonstrates

Table 3 | Pipeline stages and their acceptance evidence.

Stage	Acceptance evidence
Literature to tool	Existing Paper2Agent conversion upgraded; four MCP tools retained, with call traces
Independent verification	Decimal definitions over represented inputs; float32/64 dtype preservation
Falsification	Collapsed target and stationary-noise controls must be rejected
Repeatable delivery	82 collapse and 14 cancellation cases feed the site, JSON, CSV and this report
Reuse	NumPy kernels, optional first-order Torch kernels and a numerical audit in CI
Provenance	Locked dependencies, source SHA-256 and a checksummed source bundle

Reuse and limits

Use `framework.stable_numerics` for NumPy and `framework.stable_numerics_torch` for optional Torch. The existing MCP/agent retains four tools and explicitly separates teacher logits from legacy probabilities. Supported inputs are finite float32/float64; exponent underflow and already-rounded inputs cannot be recovered. Torch custom loss backward is first-order only; AMP/bfloat16 and higher-order derivatives need separate qualification. Static screening is heuristic. A passing built-in kernel does not certify a consumer's untested backend.

Reproduce

```
uv sync --locked --extra dev
uv run --no-sync python reproduce.py
uv run --no-sync python -m pytest tests -q
```

The ZIP includes source, lock, tests, evidence, and checksums. JSON and CSV expose every displayed case.

Optimizer-level before / after

A corrected mechanistic rerun uses trained modular-addition MLP teachers and three declared seeds. Both students start with identical represented logits; each runs 2000 full-batch Adam steps.

Table 4 | Final displacement and teacher held-out accuracy for every seed pair.

Seed	Standard final parameter L2 displacement	Fixed final parameter L2 displacement	Teacher held-out accuracy (%)
0	0.17216121	0	0.07117
1	0.15021692	0	0
2	0.18160596	0	0

All teachers reached 100% training accuracy but approximately 0-0.0712% held-out accuracy. This is a stationarity mechanism experiment, not evidence for generalisation or the full paper's training results. Saved weights independently reproduce the primary endpoint. All seeds, split identities, protocol and model checkpoints are included in the source bundle. The first run failed because a full-batch teacher cache differed from a training-batch forward. The failed code, evidence and checkpoints are preserved; the corrected run asserts exact initial logit equality.

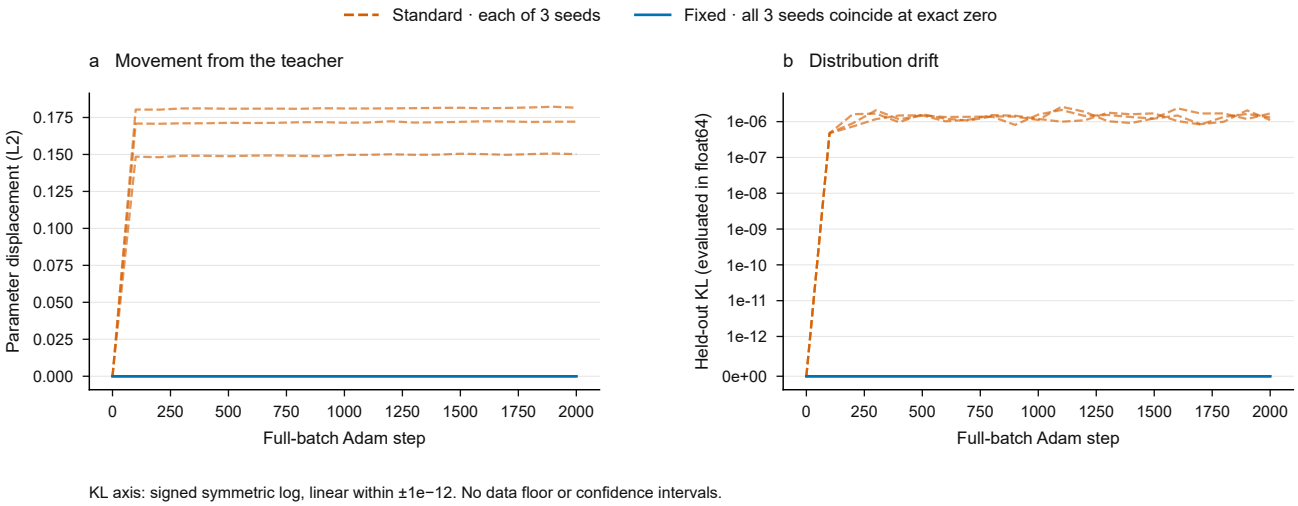


Figure 2 | Stationarity across all three seed pairs. a, Parameter L2 displacement from the initial teacher. b, Held-out KL evaluated in float64. Every line is one paired training trajectory; all fixed trajectories coincide at zero. No error bars or population inference. The KL axis is symmetric log with a linear region within plus or minus 1e-12; signed numerical residuals are retained.

Data availability

Original protocols, raw measurements, seed/split identifiers and saved-model artifacts are available in the evidence record and the checksummed reproduction bundle. Every specified outcome is retained. The repository records code, data and environment hashes; artifact acceptance can be rerun independently.

Code availability

Source and tests are included in `reproduce.zip` and in the EIDOS repository. Follow the bundle README for CPU acceptance and the recorded runtime for retraining. The original numerical method is attributed to its authors in the references.

Reporting and submission metadata

This is a technical validation report. Authors, affiliations, funding, author contributions and competing-interest declarations must be completed by the human research team before journal submission. No inferred personal or funding metadata has been supplied. Technical production follows the linked figure and manuscript guides; scientific suitability and editorial acceptance require separate review.

References

1. Barron, J. T., Gnanapragasam, D. & Tschannen, M. *A Fix for Softmax and Cross-Entropy*, version 3. arXiv:2610.31642v3.
2. Nature. Research figure guide.
3. Nature. Formatting guide.